CHEM 105 Workshop 1

Station #1

A Wellesley student's nutritious breakfast…

The United States Recommended Daily Allowance of iron is 18mg. Iron-fortified cereals often contain large amounts of iron to help people meet this dietary need (the demonstration in front of you is the extracted iron from a box of cereal). Suppose you eat 2 servings of one of the cereals in front of you for breakfast (choose one and write down which one you chose). How many iron atoms have you just ingested?

Now suppose you had to eat each iron atom one at a time, at a rate of one atom per second. How long would it take to ingest the above number of iron atoms? (For comparison, the age of the universe is 13.7 billion years, according to the NASA Wilkinson Microwave Anisotropy Probe.)

People generally have approximately 3g of iron in their whole body. Most of the iron in the body is used to transport oxygen (O_2) through the bloodstream so oxygen can reach every cell in the body. One iron binds to one oxygen molecule (this process is mediated by a protein called hemoglobin, which you may have heard of). Assume that about 80% of the iron in your body is involved in transporting oxygen at any given time. How many *grams* of oxygen are being transported in your bloodstream at this moment?

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Station #2:

A contest of sorts…

In front of you are 5.0 g of each of the following substances:

water (H_2O)
ethanol (C_2H_5OH)
sucrose ($C_{12}H_{22}O_{11}$)
caffeine ($C_8H_{10}N_4O_2$)

A) Without referring to your calculator, determine which vial contains the largest number of molecules.
B) Which vial contains the largest number of *atoms*? This time you will need your calculator.
C) By looking at the two substances, which liquid – ethanol or water – is denser?

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Station #3:

Are you a "black diamond" stoichiometry problem-solver?

→Beginner's "slope":

1) How many atoms of hydrogen are in one molecule of water?
2) How many molecules of water are in 1.5 moles of water?
3) What is the mass, in grams, of 1.00 mol of each of the following substances:
   a. Sulfur (S) b. Sodium Chloride (NaCl)
4) Using dimensional analysis (*also known as unit analysis*), calculate the total number of molecules in 2.58 grams of glucose, $C_6H_{12}O_6$.
5) Using dimensional analysis, calculate the molarity of a 2.00 L solution that contains 223.5 grams of $CuBr_2$.
6) Using dimensional analysis, calculate how many moles of NaOH are contained in 0.50 L of a 0.065 M NaOH solution.

→Intermediate "slope":

7) Using dimensional analysis (also known as unit analysis), calculate the total number of atoms in 0.260 mol of glucose, $C_6H_{12}O_6$.
8) Using dimensional analysis, determine the mass in grams of 1.45 mol $(NH_4)_2CO_3$.
9) 50.0 mL of an aqueous NaCl solution yielded 1.25 grams NaCl on evaporation. What was the molarity of the original solution?
10) Balance the following reaction: $C_9H_{10}O + O_2 \rightarrow CO_2 + H_2O$
    a. Using dimensional analysis, determine how many moles of oxygen would be required to produce 0.145 moles of carbon dioxide.
11) A substance was found to be composed of 22.9 grams of Na, 21.5 grams of B, and 55.7 grams of O. What is the empirical formula of this compound?

→Advanced "slope":

12) In *dilute* nitric acid, HNO_3, copper metal dissolves according to the following equation:
    $3Cu_{(s)} + 8HNO_{3(aq)} \text{-----> } 3Cu(NO_3)_{2(aq)} + 2NO_{(g)} + 4H_2O_{(aq)}$
    a. Using dimensional analysis, calculate how many molecules of HNO_3 are needed to dissolve 11.45g of Cu?
    b. If the reaction required 682 mL of dilute nitric acid, what was the molarity of the nitric acid solution? (Use dimensional analysis to solve this problem.)
13) Calculate the percentage composition by mass of each element in $(CH_3)_2CO$.
14) When 0.684 g of an organic compound containing only C, H, and O was burned in oxygen 1.312g CO_2 and 0.805g H_2O were obtained. What is the empirical formula of the compound?
15) Phencyclidine is $C_{17}H_{25}N$. A sample suspected of being this illicit drug was found to have a percentage composition of 83.71% C, 10.42% H, and 5.61% N. Does this data acceptably match the theoretical data for phencyclidine?

(Answers to the above problems are on the back.)

To get more practice, search the web for any "stoichiometry practice problems with answers."

If you practice stoichiometry problems enough, you might become…
…an OLYMPIC STOICHIOMETRIC PROBLEM SOLVER! WAHOO!

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Answers to Station #3 Problems:

1) 2 Hydrogen atoms → H_2O
2) $9.03x10^{23}$ molecules H_2O
3) a. 32.1 grams/mole S b. 58.4 g/mol NaCl
4) $8.63x10^{21}$ glucose molecules
5) 0.500 mols $CuBr_2$/L solution
6) 0.032 moles NaOH
7) $3.76x10^{24}$ atoms
8) 139g $(NH_4)_2CO_3$
9) 0.428M NaCl
10) $1C_9H_{10}O + 11O_2 \rightarrow 9CO_2 + 5H_2O$ a. 0.177 moles O_2
11) $Na_2B_4O_7$
12) a. 2.89x1023 molecules HNO_3 b.0.704M HNO_3
13) 62.0% C, 10.4% H, 27.6% O
14) C_2H_6O
15) Theoretical data (83.89% C, 10.35% H, 5.76% N) are consistent with experimental results.